

Skill Pods

Complete Project Specification & Master Defense Dossier

A comprehensive end-to-end blueprint detailing project objectives, problem statement mapping, core workflow, system architecture, database schemas, and plain-English answers to all technical viva questions.

Project Name	SkillPods (Demand-First Experiential Engineering Operating System)
Creator & Role	Sanket Bhende (Lead Architect & SuperAdmin • Computer Engineering)
Problem Statement	SIH26044 (Primary: Academia-Industry Skill Mapping & Placements) • SIH26043 (Secondary)
Target Audience	Engineering Students, Indian SMEs/Startups, Industry Mentors, College Placement Cells
Live Production URL	https://skill-pods.pages.dev
GitHub Repository	https://github.com/Sakshi-patil48/Skill-Pods.git

1. ■ Objective & Problem Statement

Main Objective: To eliminate throwaway college capstone projects and bridge the gap between engineering colleges and Indian Small & Medium Enterprises (SMEs), allowing students to solve real commercial problems, earn milestone stipends in escrow, and graduate with verified cryptographic Skill Passports for campus placements.

What Issue it Solves:

- **For Students:** Lack of real-world software experience and unverified resumes that recruiters cannot trust.
- **For SMEs:** Inability to afford expensive dev agencies (charging ₹5,00,000+ for basic software MVPs).
- **For Colleges:** Difficulty in tracking verifiable student internships and generating NAAC/NIRF industry reports.

2. ■ Target Audience & Stakeholders

Stakeholder	Who They Are	Why They Use SkillPods
Students	Engineering / CS college students	Gain authentic industry experience, earn sprint bounties in escrow, and earn a QR-verified Skill Passport.
SMEs / Startups	Small & Medium Enterprises	Get mentored custom software delivered at 70% lower cost than traditional IT dev agencies.
Industry Mentors	Staff / Senior Software Engineers	Guide student teams, enforce code quality, and scout top engineering talent before campus placements.
Colleges / Deans	Universities & Placement Cells	Track verified student placement metrics and export 1-click NAAC (Criterion 3.5.1/5.2.1) & NIRF reports.

3. ■ Core Platform Features

- **Demand-First SME Problem Hub:** SMEs post real business challenges with escrow milestone budgets.
- **AI Pod Match Engine:** Algorithmic matching assembling 3-student pods (Frontend, Backend, AI/Data) with 96% fit scoring.
- **Smart Escrow Milestone Vault:** Safe milestone fund locking; auto-released to students upon mentor approval.
- **Industry Mentor Sprint Gates:** Senior engineers review pull requests and test coverage before approving milestones.
- **Cryptographic Skill Passport (HMAC-SHA256):** Unforgeable proof-of-work stamped with GitHub commits & QR codes.
- **Academic Document Vault:** Verified university marksheets, CGPA, and certificate lightbox.
- **LinkedIn-Style Community & Direct Messaging:** Professional cross-role directory and private 1-on-1 chat streams.

4. ■■ Tech Stack & 4-Tier Architecture

Layer	Technologies Used	Why This Stack Was Chosen (Simple Rationale)
Frontend	React 19, TypeScript, Tailwind CSS v4, Motion, WebGL	Super-fast edge rendering, modern dark UI, smooth animations, zero runtime null errors.
Backend	Node.js, Express.js, REST APIs, WebSockets, WebRTC	Asynchronous event-driven I/O handling logins, live messaging, and low-latency sprint room streams.
AI Engine	Google Gemini API, GURU Co-Pilot, Web Speech API	Converts voice to PRDs and automatically breaks problems into 5 developer Kanban tasks.
Database	MongoDB Atlas Cloud, WiredTiger Snappy Engine	Flexible document model storing nested pod rosters, skill passports, and private chat streams.
Security	HMAC-SHA256, PBKDF2 (SHA-512), Sliding Rate Limiter	Tamper-proof passport integrity, brute-force defense, and timing-safe equality comparison.

5. ■ Step-by-Step System Workflow (5 Simple Steps)

- SME Posts Requirement:** SME posts problem and deposits milestone funds (e.g. ■25,000) into the Smart Escrow Vault.
- AI Assembles Skill Pod:** System matches a 3-student team (Frontend, Backend, AI/Data) based on skill synergy.
- Students Build in Sprints:** Pod develops solution on GitHub with continuous testing and commit telemetry.
- Mentor Reviews & Approves Gate:** Industry mentor audits the pull request code quality and test pass rates.
- Payout & Passport Issued:** Escrow auto-disburses funds via UPI; student receives a QR-verified Skill Passport.

6. ■ Data Requirements, Security & Deployment

- **Data Inputs & Outputs:** Takes SME problem specs & student code; produces verified production apps, passports & NAAC reports.
- **Cloud Storage:** MongoDB Atlas multi-collection store (users, pods, skillPassports, messages, securityLogs).
- **Authentication & Password Defense:** Node.js crypto PBKDF2 (SHA-512) with randomized 32-byte salt hashing.
- **DDoS & Injection Defense:** Sliding-window rate limiter (120 req / 15 min) + recursive NoSQL operator sanitizer.
- **Hosting & Scaling:** Frontend globally edge-cached on Cloudflare Pages; containerized with Docker for horizontal scaling.
- **Automated Testing:** Vitest test runner executing 10/10 unit & security test assertions in ~550ms with 0 TypeScript errors.

7. ■■■ Technical Challenges & Master Viva Defense (Cheat Sheet)

Q1: What are the main technical challenges in building SkillPods? Three main challenges: (1) State Synchronization between student PRs, mentor approvals, and escrow releases; (2) Tamper-Proof Verification ensuring nobody can forge a Skill Passport; and (3) Low-Latency Telemetry for real-time collaboration. We solved these using HMAC-SHA256 signatures, sliding-window rate limiters, and Cloudflare edge caching.
Q2: How does each core feature work internally? • AI Matching calculates a 0-100% score from tech stack overlap, commit speed, and test quality. • Escrow locks money in a smart vault and releases it only when the mentor clicks approve. • Skill Passport generates a SHA-256 hash connecting the student ID, mentor ID, and merged GitHub commit. • Document Vault stores encrypted PDF marksheets and verified CGPA records for placement cells.
Q3: What are the key dependencies? React 19 (Frontend), Node.js/Express (API Gateway), MongoDB Atlas Cloud (Database), Google Gemini API (AI PRD generation), and Cloudflare Pages (Edge CDN).
Q4: How can performance be optimized? By utilizing Cloudflare Edge CDN caching (sub-20ms latency), MongoDB WiredTiger Snappy compression, GPU-accelerated CSS transforms, and code-splitting large bundles.
Q5: What are the best practices implemented for this stack? Strict TypeScript type safety, constant-time hash equality (crypto.timingSafeEqual), modular Express routing, OWASP rate limiting, and 100% pre-funded escrow gating.
Q6: How does SkillPods protect against SME payment defaults? SMEs must deposit milestone funds into the Smart Escrow Vault before the sprint starts. Once the Industry Mentor approves the code and CI/CD tests pass ≥95%, the funds are automatically disbursed to students without requiring further SME approval.